

// PROJECT LUNAR LANDER

Neuroevolutionary Precision Pilot

Training Neural Networks using Genetic Algorithms for Autonomous Landing

// THE CHALLENGE: LUNAR LANDING

The mission requires a software pilot to navigate a lunar module from the top center to a specific landing pad at coordinates (0,0) between two yellow flags.

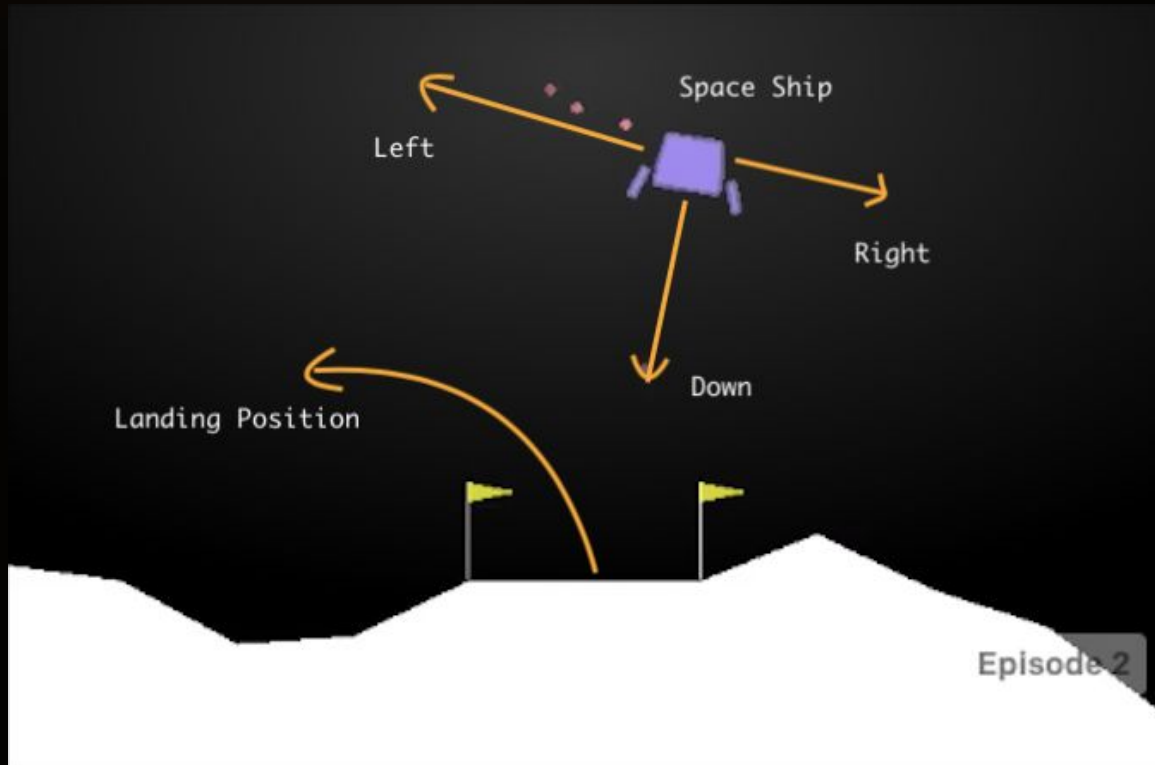
- Precision touchdown between flags
- Minimal fuel consumption
- Zero-crash safety tolerance

They used CGI to fake the moon landing

CGI in 1996:



// GYMNASIUM ENVIRONMENT



LunarLander-v3

A classic rocket trajectory optimization problem. It uses **Box2D** physics to simulate gravity, momentum, and thrust interactions.

$$F_{\text{thrust}} + F_{\text{gravity}} = m \cdot a$$

The agent must balance vertical descent and horizontal tilt simultaneously to avoid a catastrophic impact.

// WHAT THE PILOT SEES

Position

Coordinates (X, Y)
relative to the landing
pad center.

Orientation

Angle of the craft and
its angular velocity
(spin).

Velocity

Linear velocity vectors
in both horizontal and
vertical axes.

Leg Contact

Binary sensors for
both left and right
landing gear contact.

// CONTROLS & EXECUTION

0: Idle

Engines off. Saves fuel and allows for gravity-assisted descent.

1: Left Thruster

Fire side engine to rotate clockwise and move left.

2: Main Engine

Maximum thrust to counteract gravity and slow descent.

3: Right Thruster

Fire side engine to rotate counter-clockwise and move right.

// THE BRAIN: ARCHITECTURE

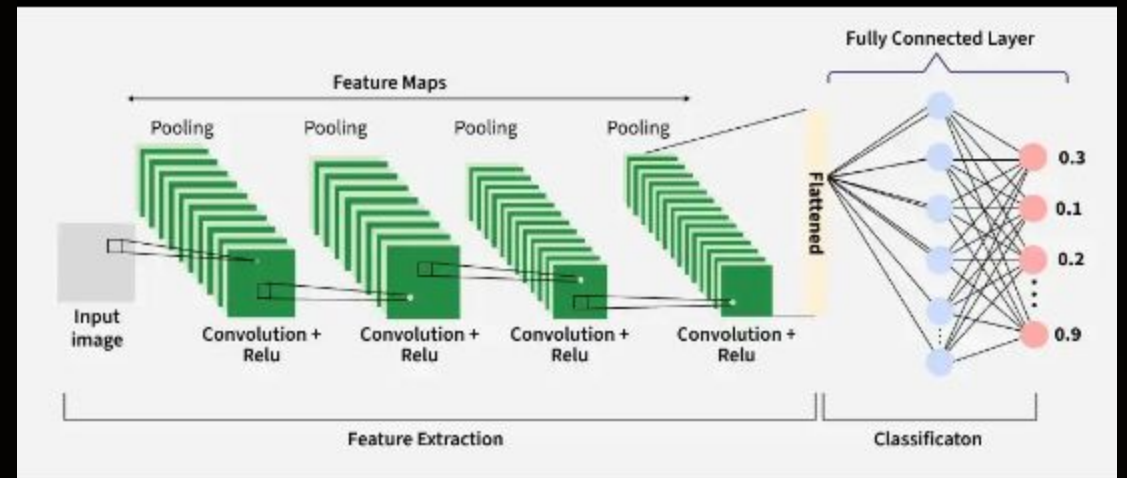
MLP Structure: $8 \rightarrow 64 \rightarrow 32 \rightarrow 4$

A Multi-Layer Perceptron acts as the decision-making policy. It maps 8 sensory inputs directly to 4 action probabilities.

Activation: Tanh (Hidden), Argmax (Output)

Complexity: 2,724 Total Genes (Weights + Biases)

Inference: Fast matrix multiplication via NumPy



// METHODOLOGY: NEUROEVOLUTION

Genotype Encoding

Neural network weights are flattened into a single array (chromosome). We evolve the weights instead of using Gradient Descent.

Two-Point Crossover

Preserves blocks of correlated weights, allowing for more stable trait inheritance compared to random shuffling.

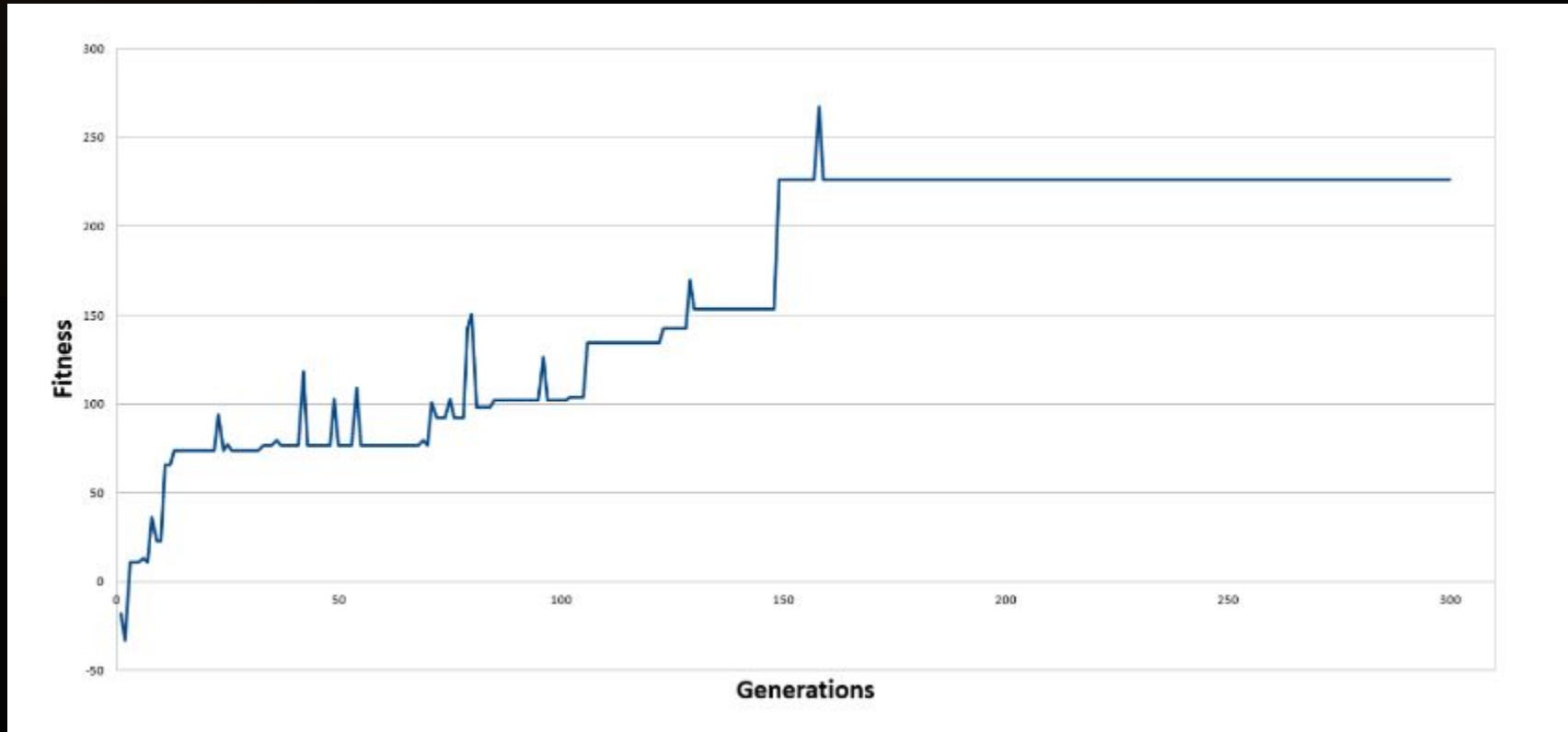
Selection Process

Tournament Selection (K=6): Picks the strongest candidates from small groups to be parents of the next generation.

Top-6 Elitism

Ensures that the best pilots are never lost, preserving progress across generations unconditionally.

// TRAINING PROGRESS



// REWARD SHAPING

Precision Bonus (+80): Fires when both legs touch. Incentivizes landing right on $x=0$ center.

Soft Landing Bonus (+40): Penalizes high vertical velocity. Teaches the agent to hover gently.

Upright Bonus (+30): Rewards landing at a perfect 0° angle. Eliminates side-scrapers.

Exploit Prevention: Removed "Touch-and-Go" hacks by averaging fitness over 5 random episodes.



// ESCAPING LOCAL OPTIMA

Adaptive Mutation

Weak solutions mutate heavily (25%) to explore new strategies, while elite solutions mutate lightly (3%) to fine-tune successful traits.

STAGNATION LIMIT: 25

If fitness plateaus for 25 generations, mutation is boosted by 1.4x to "kick" the population out of the local optimum.

```
90 # =====
91 # 3. FITNESS FUNCTION
92 #
93 # The environment's built-in reward already heavily penalises:
94 #   • crashing (-100)
95 #   • going off-screen
96 #   • fuel use
97 # and rewards:
98 #   • landing between flags (+100..+140)
99 #   • both legs touching (+10 each)
100 #
101 # We ADD three custom shaped bonuses to guide evolution:
102 #
103 # A) PRECISION BONUS - rewards being close to x=0 at landing.
104 #   Gives up to +80 extra for a centred touchdown.
105 #
106 # B) SOFT LANDING BONUS - rewards low vertical speed at contact.
107 #   Penalises hard crashes that the env already punishes.
108 #
109 # C) UPRIGHT BONUS - rewards small angle at touchdown.
110 #   Prevents landers that arrive sideways.
111 #
112 # We REMOVE the "touch-and-go" hack from the original which
113 # taught the agent to tap the ground and fly away instead of
114 # actually landing.
115 # =====
116 EVAL_EPISODES = 5 # more episodes = more stable fitness signal
117 MAX_STEPS = 600 # hard cap - prevents infinite hover
118
```

Safe Landing Achieved through Neuroevolution

Questions?